

Victoria Vehicle Market Analysis: Business Analytics Project Brief

Background

Through its Open Data Hub, the Department of Transport and Planning (Victoria State Government) provides free public access to aggregated historical vehicle registration and transfer data. The data is released as monthly CSV files recording the make, model, colour, and year of manufacture for every vehicle event.

You have been engaged by a client who operates multiple car dealerships in Queensland and wants to expand into Victoria, primarily Melbourne. Your task is to act as their business analytics consultant.

The Data

Your client has collated vehicle registration and transfer data from the Victorian open data portal covering May 2023 to June 2026. The dataset consists of two files:

1. **monthly_new_vehicle_registration_may2023_2026.csv**: new vehicle registrations
2. **monthly_vehicle_transfers_may2023_2026.csv**: used vehicle transfers (change of ownership)

Both files contain the same six variables:

Data Dictionary

Variable	Name	Type	Description
CD_MAKE_VEH	Vehicle Make	CHAR	This is a vehicle brand, for example "HOLDEN"
CD_MODEL_VEH	Vehicle Model	CHAR	The Model describes a type of vehicle make, for example 'COROLLA'
CD_CLR_BDY_VEH_P	Vehicle Colour	CHAR	Colour is the visual perception based on the electromagnetic spectrum. The colour defines shades of the object, for example "Green", "Red", etc.
NB_YEAR_MFC_VEH	Year of Manufacture	INTEGER	Year the vehicle was manufactured
TOTAL	Total number	INTEGER	Number of vehicles registered/transferred in that month for that make/model/colour/year combination
data_month_year	Month and Year	INTEGER	Month and year of vehicle registration/ transfer (YYYYMM)

Note on data quality: Make, Model, and Colour may contain abbreviated codes rather than full values, and the same vehicle may appear under multiple spellings. You will need to clean and standardise these before analysis. For example, the row

MERC B	A200	BLK
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translates to Mercedes-Benz, A200, Black.

Your Task

Explore the dataset and assess the information it captures. Then, define a business problem or opportunity that can be addressed using this dataset, and develop a comprehensive business analytics project to solve it. For example, you may investigate questions such as:

Which makes and models should the client prioritise stocking, based on registration and transfer volumes and trends?

Should the new Victorian dealerships focus their business on new vehicle sales or the used vehicle market, and in which segments?

How does the age profile of transferred vehicles differ across brands, and what does this imply about used-stock acquisition strategy?

When defining your business problem, ensure that it is clearly focused, can be answered using the available data, and provides insights that support decision-making and problem-solving.